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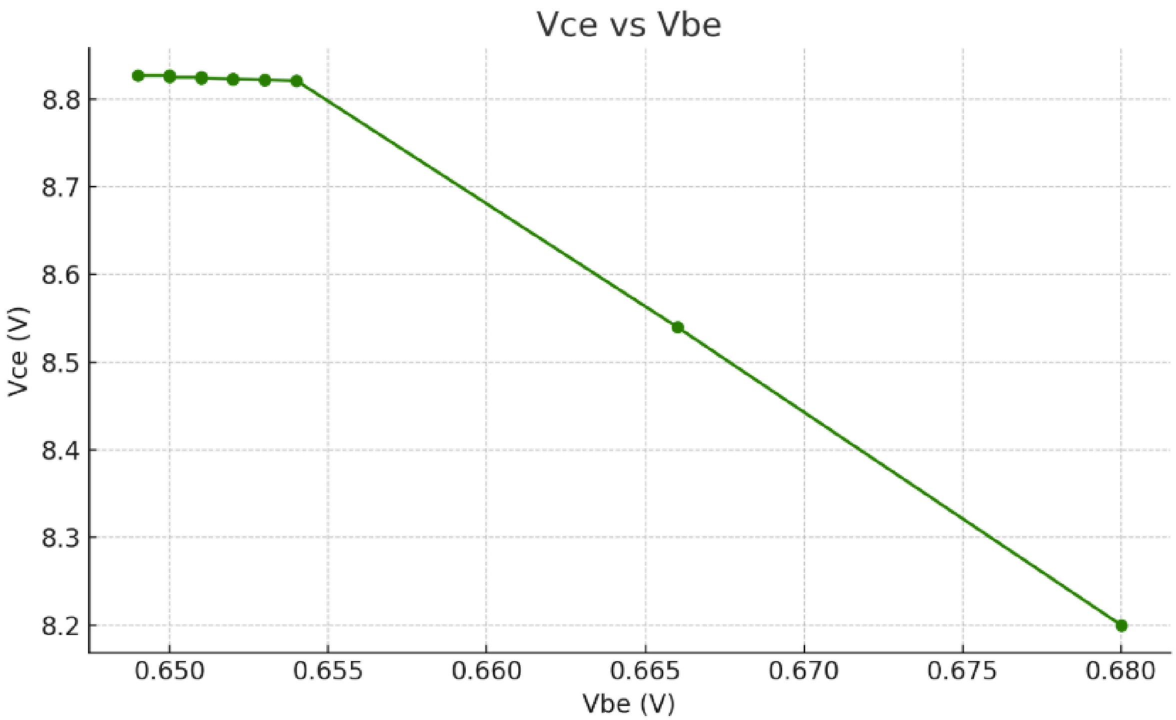
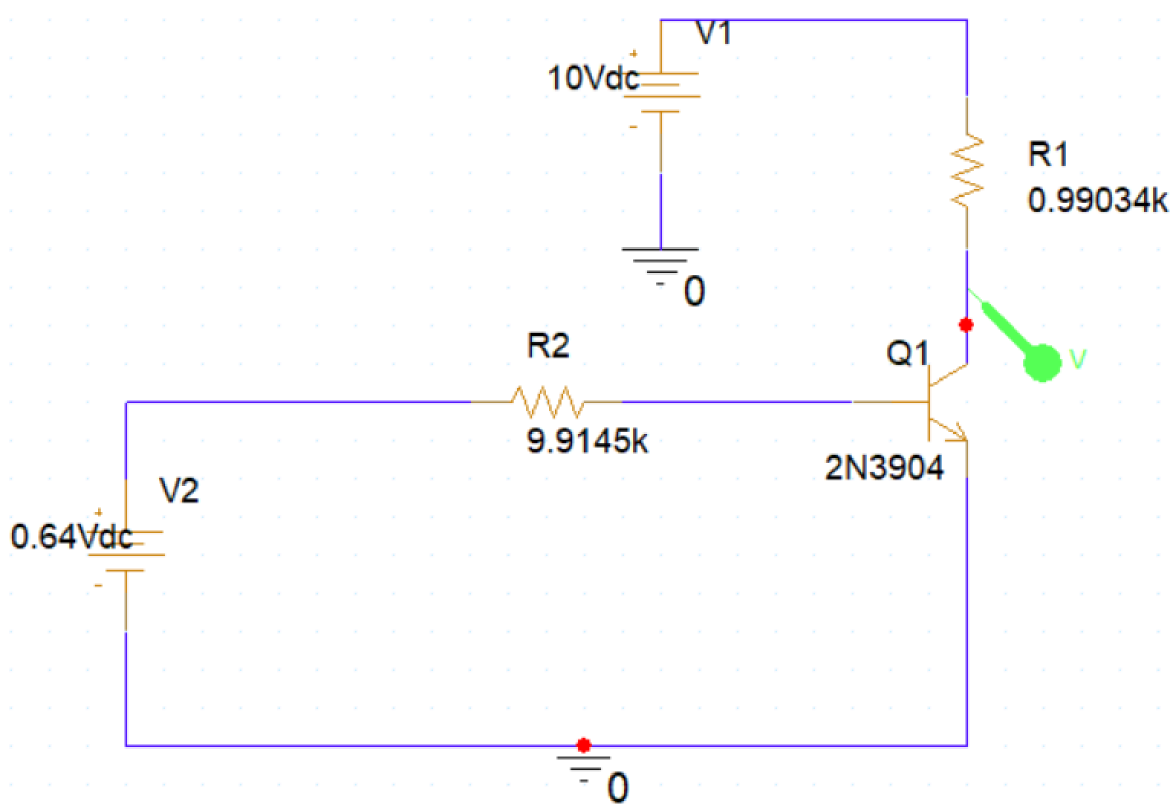
ECE 121 Lab 8

Experiment #1(Lab+Simulation):

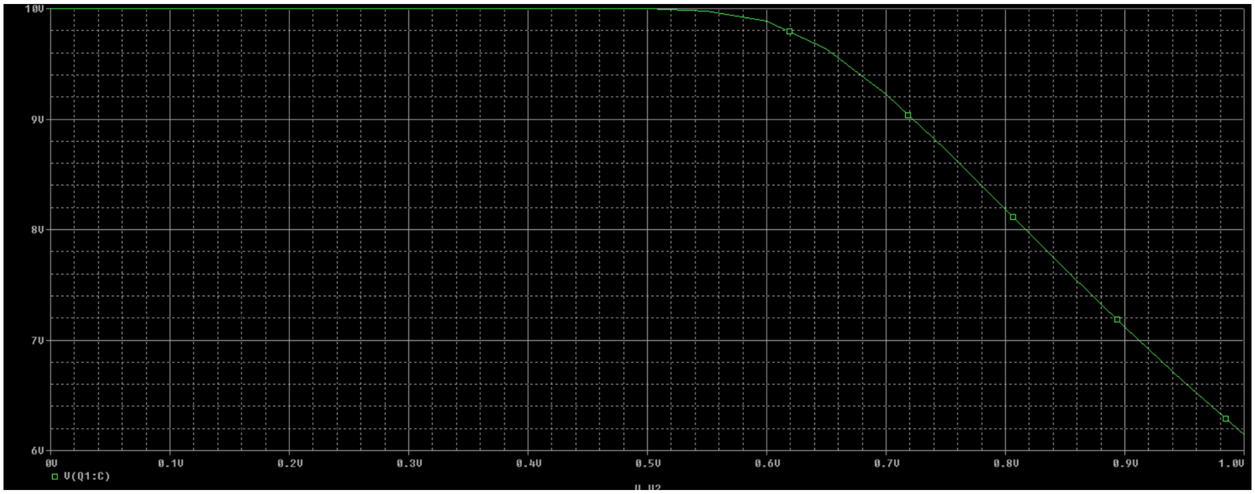
Experiment 1 Measured Values				
IB(micro Amps)	Vs	Ic	Vbe	Vce
0	0.64V	1.305mA	0.649V	8.827V
1	0.65V	1.311mA	0.650V	8.827V
5	0.68V	1.311mA	0.650V	8.827V
10	0.74V	1.311mA	0.650V	8.826V
20	0.85V	1.312mA	0.650V	8.825V
30	0.93V	1.313mA	0.651V	8.825V
40	1.04V	1.313mA	0.651V	8.824V
50	1.13V	1.314mA	0.652V	8.823V
60	1.23V	1.315mA	0.652V	8.823V
70	1.34V	1.315mA	0.652V	8.822V
80	1.45V	1.316mA	0.653V	8.821V
90	1.53V	1.317mA	0.653V	8.821V
100	1.63V	1.317mA	0.653V	8.820V
500	5.50V	1.599mA	0.666V	8.540V
1000	10.9V	3.20mA	0.68V	8.20V



Vce vs Vbe
8.8
8.7
8.6
8.5
8.4
8.3
8.2
Vce (V)
0.650
0.655
0.660
0.665
0.670
0.675
0.680
Vbe (V)



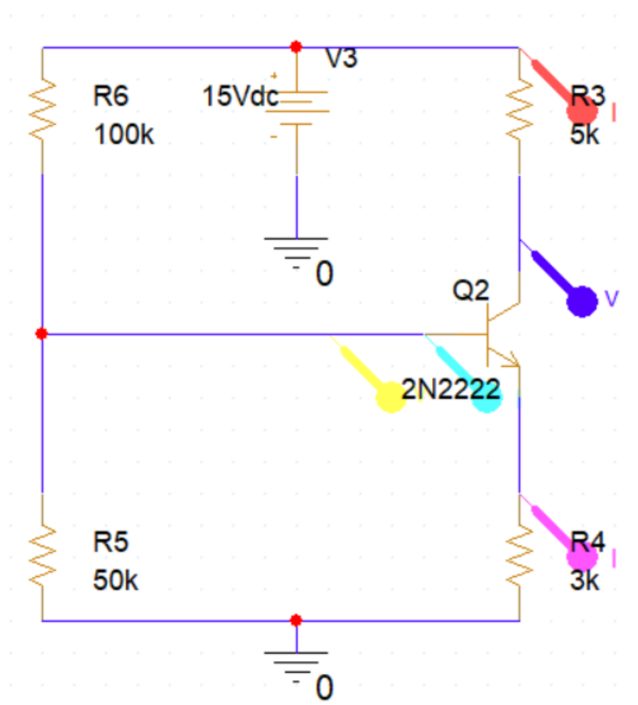
V1
10Vdc
R1
0.99034k
0
R2
9.9145k
Q1
2N3904
V
V2
0.64Vdc
0

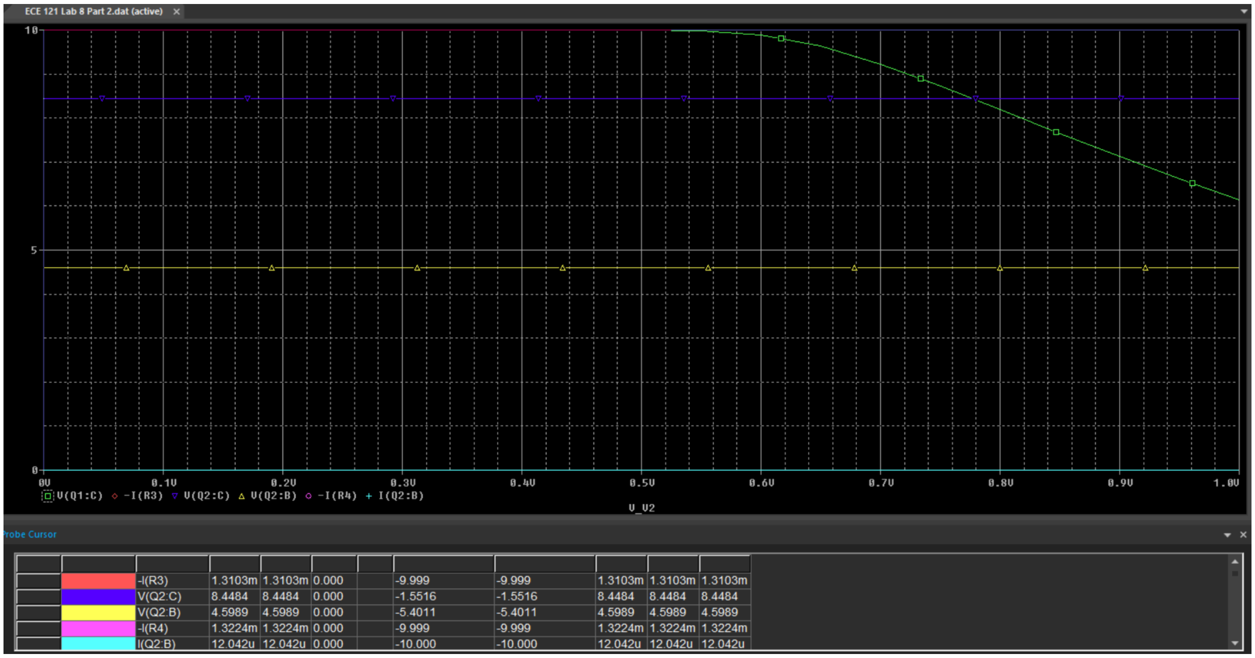


VCE vs VBB

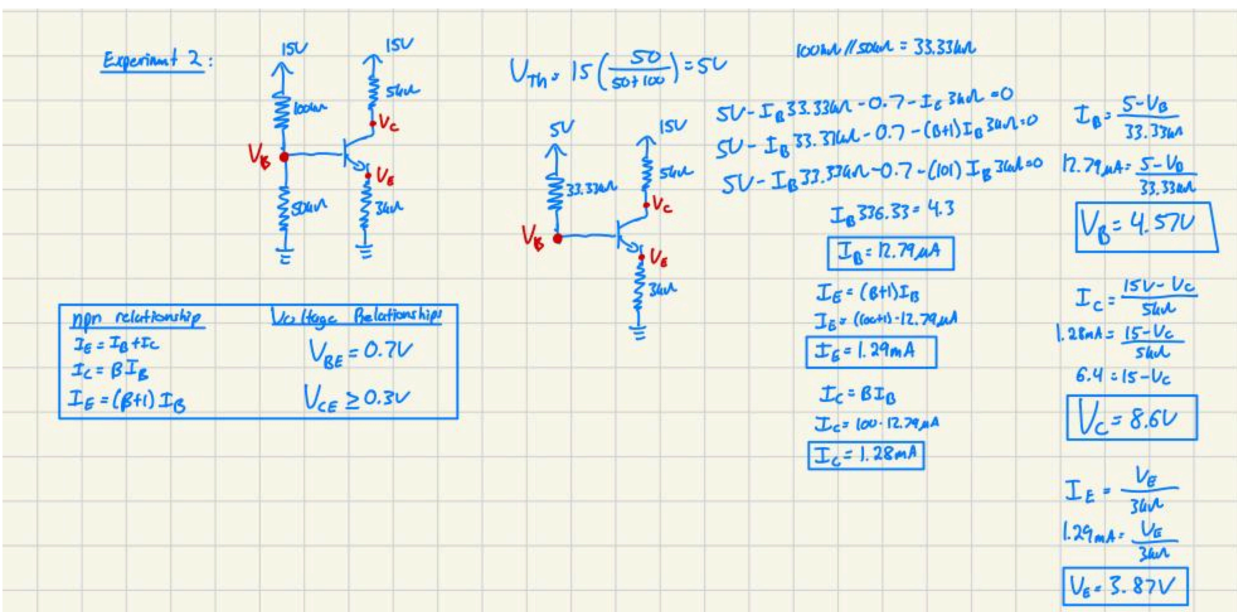
Conclusion: In the measured part of the experiment that graph is closely related to the simulated portion. Thus, I conclude that this experiment is a success due to the similarities in the graphs.

Experiment #2(Simulation+Calculation)



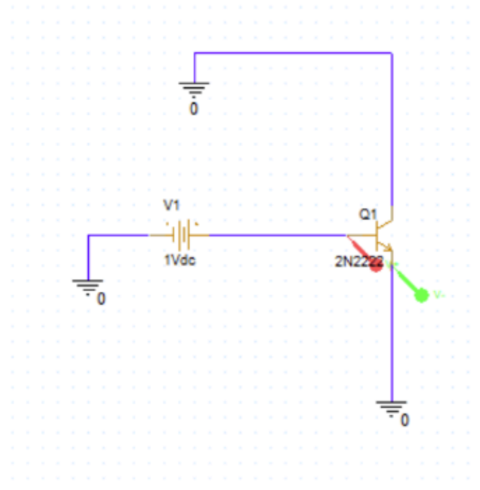


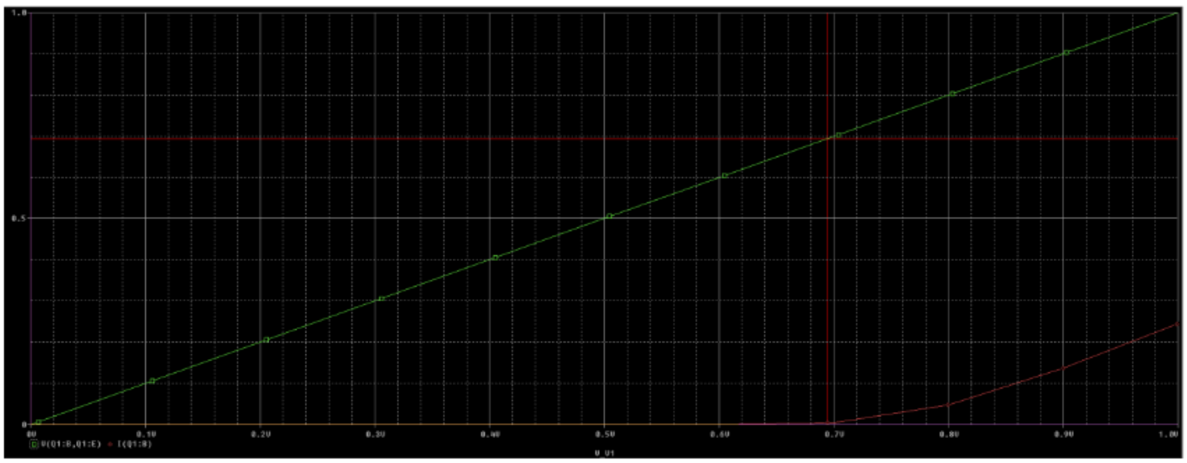
- IB=12.042uA
- IC=1.3103mA
- IE=1.3324mA
- Vbe=4.5989V
- Vce=8.4484V



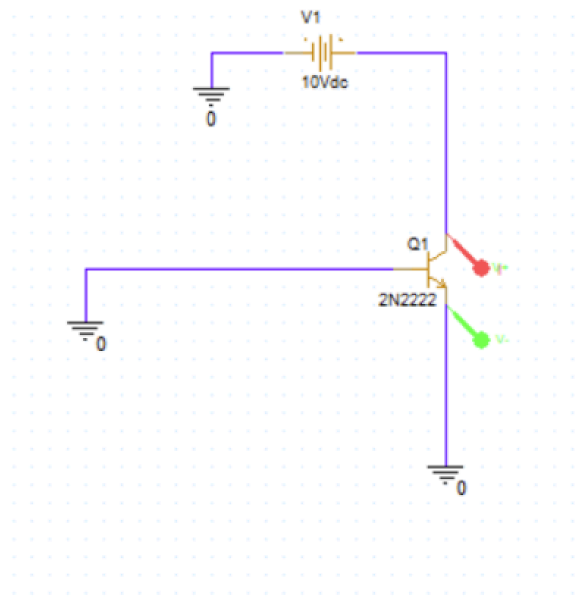
<u>Conclusion:</u> In the simulation and the calculated portion the values are very similar but there's a small difference. The beta value for simulation must be different from B=100 that I used for the calculations. That's why there's a small difference in the values, but I do believe this experiment was a success.

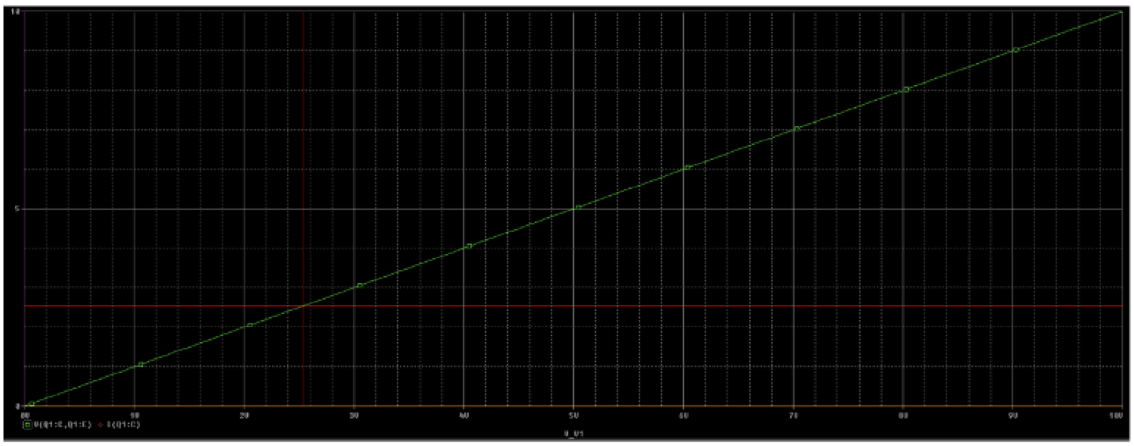
<u>Experiment #3(Simulation):</u>





	Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cursor2)		693.923m		
		X Values	693.923m	0.000	693.923m		Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
	CURSOR 1,2	V(Q1:B,Q1:E)	693.923m	0.000	693.923m		0.000	0.000	693.923m	0.000	346.961m
		I(Q1:B)	4.7476m	249.0E-24	4.7476m		-689.175m	249.0E-24	4.7476m	249.0E-24	2.3738m





	Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cursor2)		2.5359		
		X Values	2.5359	0.000	2.5359		Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
	CURSOR 1,2	V(Q1:C,Q1:E)	2.5359	0.000	2.5359		0.000	0.000	2.5359	0.000	1.2680
		I(Q1:C)	3.9195p	-1.584E-24	3.9195p		-2.5359	-1.584E-24	3.9195p	-1.584E-24	1.9598p

Conclusion: I used PSpice to make the input and output curves for the NPN transistor. The input curve showed the exponential base-emitter behavior, and the output curves showed how Ic changes with Vce at different Ib levels. In conclusion this lab helped us better understand how BJTs work.